

Yuyan Wang

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RESEARCH INTEREST

I am deeply interested in model deployment, decision learning, and human-robot collaboration within the field of **Embodied Intelligence**, focusing on how to enable intelligent agents to perform long-term planning and possess perception-reasoning-action capabilities in real physical environments. My research focuses on combining **VLA Models** and **World Models** with whole-body robot control to construct embodied systems.

EDUCATION

University of Pennsylvania MSE Computer and Information Science	2026.08 - 2028.05
University of Virginia B.A. Computer Science and Statistics (GPA: 3.97/4.0)	2022.08 - 2026.05

PUBLICATION

ICML 2026 Xuhui Kang, Sung-Wook Lee, Haolin Liu, Yuyan Wang , Yen-Ling Kuo <i>Moving Out: Physically-Grounded Human-AI Collaboration</i>	2025.07
ICRA 2026 Timur Anvar, Jeffrey Chen, Yuyan Wang , Rohan Chandra <i>Are LLMs The Way Forward? A Case Study on LLM-Guided RL for Decentralized Autonomous Driving</i>	2025.11

INTERNSHIP

Humanoid Robot Algorithm R&D Intern BXI Robotics	2026.06 - present
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- Obtain joint retargeting data for the Elf3 Humanoid Robot, train and deploy the SONIC WBC model, and achieve wholebody teleoperation and model inference for Pico VR.
- Evaluate GR00T, Pi, and LingBot-VA, train VLA to improve the Elf3's task execution capabilities, and build a VLA and SONIC control framework.
- Conducts robot data generation research based on DreamGen, using a world model to synthesize robot task video, providing high-quality data support for improving VLA post-training and generalization capabilities.

PROJECT

NASA Lunabotics Challenge <i>UVA Mechatronics and Robotics Society CS Subteam Member</i>	2023.08 - 2026.05
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- Achieved 1st place (Off World Grand Prize) out of over 60 universities in 2026 NASA Lunabotics Challenge.
- Use ROS2 Nav2 to explore SLAM-based localization and path planning methods, focusing on how to avoid obstacles such as craters to achieve autonomous movement of the robot on the lunar surface.
- Design action sequences by utilizing sensors and AprilTag for automated digging and dumping tasks.
- Establish a data transmission network between the control center (Linux) and the robot (NVIDIA Jetson).

Moving Out: Physically-Grounded Human-AI Collaboration <i>Supervisor: Yen-Ling Kuo Research Assistant</i>	2025.01 - 2026.01
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- Goal:** Enhance agents' ability to collaborate with diverse human behaviors in env with physical constraints.
- Train embodied agents using RL in the new collaborative benchmark environment, Moving Out.
- Collect and clean collaborative trajectory data, and train a Latent Dynamics Model based on this data: VAE for state encoding and decoding and MLP to learn the state transition function to predict future states.

Improvement and Optimization of Franka Robotic Arm Teleoperation <i>Supervisor: Yen-Ling Kuo Research Assistant</i>	2024.01 - 2025.04
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- Implemented teleoperation of the Franka robot arm using a Meta Quest VR controller and space mouse for diverse manipulation tasks in both physical and Gazebo simulated environments through ROS2.
- This teleoperation setup now supports 4-6 ongoing projects in the lab across multiple research directions.

王玉言

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研究兴趣

我对具身智能领域中的模型部署、决策学习、和人机协作问题充满兴趣，关注如何让智能体在真实物理环境中进行长期规划和具备感知-推理-行动的能力。我的研究聚焦于将 **VLA 模型**、**世界模型**与机器人全身控制相结合，构建能够理解高层语义目标并执行低层运动控制的具身系统。

教育背景

美国宾夕法尼亚大学 | 计算机科学硕士学位 2026.08 - 2028.05
美国弗吉尼亚大学 | 计算机科学与应用统计学士双学位 (GPA: 3.97/4.0) 2022.08 - 2026.05

论文发表

ICML 2026 Xuhui Kang, Sung-Wook Lee, Haolin Liu, **Yuyan Wang**, Yen-Ling Kuo 2025
Moving Out: Physically-Grounded Human-AI Collaboration
ICRA 2026 Timur Anvar, Jeffrey Chen, **Yuyan Wang**, Rohan Chandra 2025
Are LLMs The Way Forward? A Case Study on LLM-Guided RL for Decentralized Autonomous Driving

实习

人形机器人算法研发实习生 | 上海半醒科技有限公司 2026.06 至今

- 获得 Elf3 机器人关节映射数据，训练并部署 SONIC 全身控制模型，实现 Pico VR 全身遥操作与模型推理。
- 评估 GR00T、Pi、蚂蚁灵波等模型，后训练 VLA 提升 Elf3 任务执行能力，搭建 VLA 与 SONIC 控制框架。
- 基于 DreamGen 开展机器人数据生成研究，利用世界模型合成多样化操作轨迹，为 VLA 模型后训练和泛化能力提升提供高质量数据支持。

项目经历

NASA Lunabotics Challenge 月球机器人挑战赛 2023.08-2026.05
弗吉尼亚大学机电与机器人社团 | 计算机小组组员

- 项目成果：**代表弗吉尼亚大学参加 NASA Lunabotics Challenge 比赛，2026 年在 60 多所高校中荣获第 1 名。
- 使用 ROS2 Nav2 探索基于 SLAM 的定位与路径规划方法，重点关注如何避让陨石坑等障碍，以实现机器人在月壤表面的自主移动能力。
- 利用传感器及 AprilTag 技术设计动作序列，使机器人能够完成自动化挖掘与倾倒任务。
- 搭建控制中心 (Linux) 与机器人模块系统 (NVIDIA Jetson) 之间的网络连接与数据传输链路。

Moving Out: Physically-Grounded Human-AI Collaboration 2025.01-2026.01
导师: Yen-Ling Kuo | 科研助理

- 项目目标：**提升智能体在具有物理约束的复杂场景中对多样人类行为与未知物理属性的适应与协作能力。
- 在全新的人机协作基准环境 Moving Out 中使用强化学习训练多智能体，调试参数，比较学习效率与策略效果。
- 收集并清理协作轨迹数据，在此基础上构建 Latent Dynamics Model: 采用变分自编码器 (VAE) 进行状态编码与解码，并利用多层感知器 (MLP) 学习状态转移函数，实现对未来状态的预测。

Franka 机械臂遥控系统优化改进 2024.01 - 2025.04
导师: Yen-Ling Kuo | 科研助理

- 项目成果：**搭建的遥控系统支撑实验室 4-6 个正在进行的多个研究方向上的科研项目。
- 基于 ROS2 用 Meta Quest VR 手柄与 3D 鼠标遥控 Franka 机械臂，支持多类型实体操作任务。
- 在 Gazebo 仿真环境中搭建 Franka 机械臂遥控系统，实现与实体机械臂一致的操作体验与试验复现性。